
A COURSE

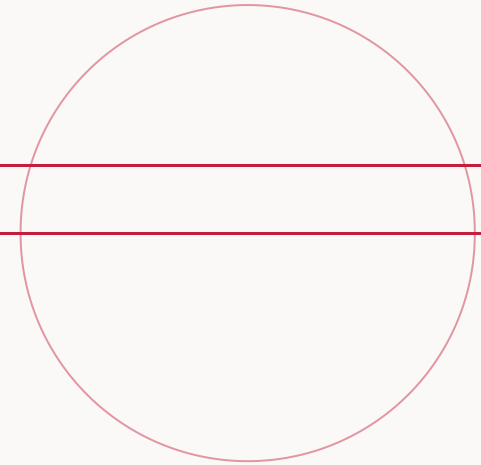
Quantitative Finance using Python

*Statistics for return & risk —
from raw prices to factor models.*

Manaranjan Pradhan

www.aithinking.in

A Course on Quantitative Finance



ABOUT THE INSTRUCTOR

Manaranjan Pradhan



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- Consulting and training on **Big Data, AI & Machine Learning**.
- An alumnus of *IIM Bangalore*.
- **20+ years** of industry experience.
- Trained **1,000+** professionals on Big Data and AI/ML.
- Adjunct faculty at *IIM Bangalore, ISB Hyderabad*, and the *Jio Institute*.

Co-author of the best-selling book **Machine Learning using Python** —
and of two Harvard Business Publishing cases: **Customer Analytics at BigBasket** and **Lead Generation at Eureka Forbes**.

COURSE OUTLINE

Eight modules, one arc

MODULE 1

Exploratory analysis

Returns and the four moments.

MODULE 2

Probability models & VaR

Fit a distribution; bound the loss.

MODULE 3

Estimation & inference

Estimates with their standard errors.

MODULE 4

Regression, β & α

The market model.

MODULE 5

Fama-French 3-factor

Size and value premia.

MODULE 6

Portfolio optimisation

Frontier, GMV, tangency.

MODULE 7

Time-series analysis

Volatility clusters, dynamic risk.

MODULE 8

Credit risk modelling

Default probability, scorecards.

*The arc is cumulative — **describe, model, infer, regress, decompose, optimise, evolve, score.***